

Data Requirements for Parameter Recovery in MAPK Signaling Dynamics Using Physics-Informed Neural Networks

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Outline

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Motivation

- MAPK signaling cascades regulate important cellular processes such as:
 - cell growth
 - proliferation
 - differentiation
 - survival
- Mathematical models can describe the dynamics of MAPK signaling.
- Accurate kinetic parameters are required to understand and predict signaling dynamics.
- Experimental measurements is:
 - limited,
 - noisy,

Why Data Availability Matters

Problem

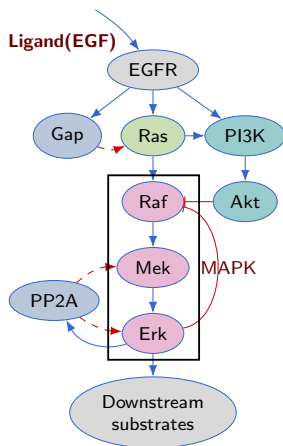
How much experimental data are actually required to reliably estimate or recover the parameters of a MAPK model?

- More observations provide more information about the dynamics.
- Too few observations may lead to inaccurate parameter estimates or recovery.

Goal

Determine the minimum number of temporal observations required for reliable MAPK parameter recovery.

MAPK Signaling Cascade



We denote

$E^* = \text{EGFR}^*$, $R^* = \text{Ras}^*$, $F^* = \text{Raf}^*$, $M^* = \text{Mek}^*$,
 $K^* = \text{Erk}^*$, $A^* = \text{AKT}^*$, $P^* = \text{PI3K}^*$, $G^* = \text{GAP}^*$,
 $S^* = \text{PP2A}^*$, $L = \text{Ligand}$, $E = \text{EGFR}$, $R = \text{Ras}$,
 $F = \text{Raf}$, $M = \text{Mek}$, $K = \text{Erk}$, $A = \text{AKT}$,
 $P = \text{PI3K}$, $G = \text{GAP}$, $S = \text{PP2A}$

- X and X^* represent the concentration levels of the **inactive forms** and **active forms** of the signaling proteins respectively.

MAPK Signaling Cascade

$$\dot{L} = -k_l L$$

$$\dot{E}^* = k_1 \frac{L(E_{Tot} - E^*)}{K_1 + (E_{Tot} - E^*)} - k_e E^*$$

$$\dot{P}^* = k_6 \frac{E^*(P_{Tot} - P^*)}{K_6 + (P_{Tot} - P^*)} + k_{12} \frac{R^*(P_{Tot} - P^*)}{K_{12} + (P_{Tot} - P^*)} - k_p P^*$$

$$\dot{A}^* = k_7 \frac{P^*(A_{Tot} - A^*)}{K_7 + (A_{Tot} - A^*)} - k_a A^*$$

$$\dot{R}^* = k_2 \frac{E^*(R_{Tot} - R^*)}{K_2 + (R_{Tot} - R^*)} - k_{10} \frac{G^* R^*}{K_{10} + R^*} - k_r R^*$$

$$\dot{F}^* = k_3 \frac{R^*(F_{Tot} - F^*)}{K_3 + (F_{Tot} - F^*)} \cdot \frac{1}{1 + K^*} \cdot \frac{1}{1 + A^*} - k_f F^*$$

$$\dot{M}^* = k_4 \frac{F^*(M_{Tot} - M^*)}{K_4 + (M_{Tot} - M^*)} - k_9 \frac{S^* M^*}{K_9 + M^*} - k_m M^*$$

$$\dot{K}^* = k_5 \frac{M^*(K_{Tot} - K^*)}{K_5 + (K_{Tot} - K^*)} - k_8 \frac{S^* K^*}{K_8 + K^*} - k_d K^*$$

$$\dot{G}^* = \frac{k_{11} E^*(G_{Tot} - G^*)}{K_{11} + (G_{Tot} - G^*)} - k_g G^*$$

$$\dot{S}^* = \frac{k_{13} K^*(S_{Tot} - S^*)}{K_{13} + (S_{Tot} - S^*)} - k_s S^*$$

Physics-Informed Neural Networks (PINNs)

PINN is a neural network that allow the incorporation of the governing equations (ODEs/PDEs) in the training process.

It approximates the MAPK state variables using a neural network:

$$\hat{\mathbf{x}}(t) = \mathcal{N}_{\phi}(t)$$

Automatic differentiation provides:

$$\frac{d\hat{\mathbf{x}}}{dt}.$$

The MAPK ODE residual is

$$\mathbf{r}(t) = \frac{d\hat{\mathbf{x}}}{dt} - \mathbf{f}(\hat{\mathbf{x}}, \boldsymbol{\theta}).$$

The PINN attempts to minimize this residual:

$$\mathbf{r}(t) \rightarrow \mathbf{0}.$$

PINN Loss Function

The total loss is

$$\mathcal{L} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{phys}}\mathcal{L}_{\text{phys}} + w_{\text{IC}}\mathcal{L}_{\text{IC}}.$$

Data loss

$$\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_{i=1}^N \left\| \hat{\mathbf{x}}(t_i) - \mathbf{x}_i^{\text{obs}} \right\|^2$$

Physics loss

$$\mathcal{L}_{\text{phys}} = \frac{1}{N_f} \sum_{j=1}^{N_f} \left\| \mathbf{r}(t_j) \right\|^2$$

Initial-condition loss

$$\mathcal{L}_{\text{IC}} = \left\| \hat{\mathbf{x}}(0) - \mathbf{x}_0 \right\|^2.$$

The experiment uses $w_{\text{data}} = 1$, $w_{\text{phys}} = 10$, and $w_{\text{IC}} = 1$

Parameter Recovery Procedure

The neural-network parameters and the unknown system's parameters are optimized simultaneously:

$$(\phi^*, \theta^*) = \underset{\phi, \theta}{\operatorname{argmin}} \mathcal{L}.$$

The MAPK kinetic parameters are divided into three groups:

- Catalytic rate constants $(k_1, k_2, \dots, k_{13})$
- Michaelis–Menten constants $(K_1, K_2, \dots, K_{13})$
- Degradation rate constants $(k_l, k_e, k_p, k_a, k_r, k_f, k_m, k_d, k_g, k_s)$

Experimental Design and Training Procedure

We considered three different observations per each state variable and three different levels of gaussian noise.

	0%	5%	10%
25 observations	✓	✓	✓
50 observations	✓	✓	✓
100 observations	✓	✓	✓

- 1 Generate the reference MAPK solution.
- 2 Generate synthetic observations.
- 3 Add the prescribed measurement noise.
- 4 Initialize the recoverable parameters 20% away from ground truth.
- 5 Train the PINN for 1600 epochs.
- 6 Select the parameter set corresponding to the minimum total loss.
- 7 Compare recovered parameters with ground truth.
- 8 Compare recovered dynamics with the reference solution.

Evaluation Metrics of Parameter Recovery

1. Training Loss

$$\mathcal{L}_{\text{total}}$$

2. Relative Parameter Error

$$\text{Relative Error (E}_{\text{rel}}) = \frac{|\theta_{\text{est}} - \theta_{\text{true}}|}{|\theta_{\text{true}}|} \times 100\%.$$

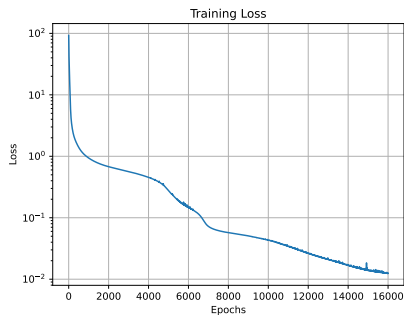
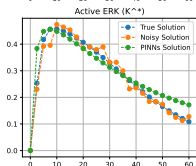
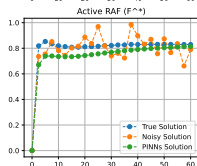
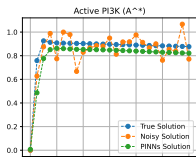
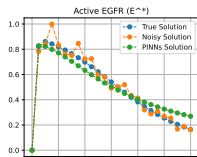
3. Trajectory Recovery

Comparison between $\mathbf{x}_{\text{true}}(t)$ and $\mathbf{x}_{\text{est}}(t)$

Parameter Recovery Results

Relative Error for Catalytic Constants Across Data Conditions

	0%	5%	10%
25	10.67%, 8.82×10^{-4}	25.91%, 6.77×10^{-2}	30.73%, 9.32×10^{-2}
50	8.40%, 2.57×10^{-4}	15.06%, 1×10^{-3}	18.76%, 3.82×10^{-3}
100	7.41%, 2.32×10^{-4}	10.54%, 1.14×10^{-3}	14.6%, 3.82×10^{-3}

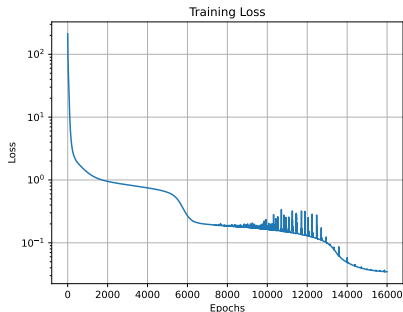
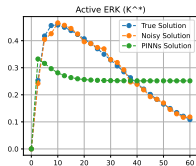
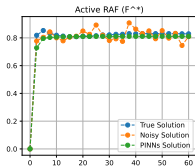
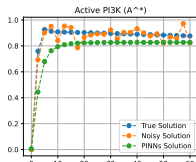
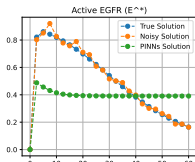


Trajectory Recovery for 25 obs. with 5% noise

Parameter Recovery Results

Relative Error for MM Constants Across Data Conditions

	0%	5%	10%
25	24.79%, 5.86×10^{-3}	57.84%, 2.54×10^{-2}	57.94%, 2.81×10^{-2}
50	15.37%, 1.15×10^{-4}	68.86%, 8×10^{-3}	71.36%, 1.14×10^{-2}
100	15.81%, 1.02×10^{-4}	20.64%, 1.11×10^{-3}	21.33%, 4.21×10^{-3}

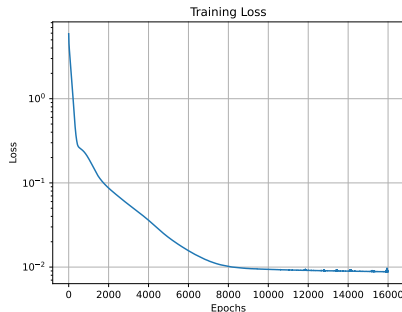
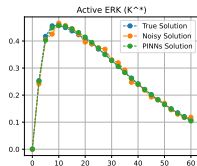
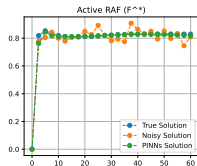
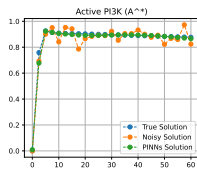
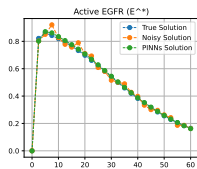


Trajectory Recovery for 25 obs. with 5% noise

Parameter Recovery Results

Relative Error for Degradation Constants Across Data Conditions

	0%	5%	10%
25 obs	3.04%, 3.99×10^{-5}	5.26%, 7.77×10^{-4}	10.31%, 3×10^{-3}
50 obs	1.74%, 8.69×10^{-5}	5.41%, 8.55×10^{-2}	6.94%, 3.29×10^{-2}
100 obs	5.63%, 8.04×10^{-5}	5.53%, 9.46×10^{-2}	5.44%, 3.64×10^{-2}



Trajectory Recovery for 25 obs. with 5% noise

Minimum number of observations required for reliable parameter recovery

Let $E_{\max} = 20\%$ be the maximum acceptable relative parameter error. An observation size is considered sufficient when $E_{\text{rel}} \leq E_{\max}$. Therefore,

$$N_{\min} = \min \{N_{\text{obs}} : E_{\text{rel}} \leq E_{\max}\}.$$

Parameter Group	0% Noise	5% Noise	10% Noise
Catalytic	25 obs	50 obs	50 obs
Michaelis–Menten	50 obs	100 obs	100 obs
Degradation	25 obs	25 obs	25 obs

Main Observations

- Parameter recovery accuracy changes with the number of temporal observations.
- Measurement noise affects the accuracy of parameter recovery.
- Parameter-recovery accuracy differs among parameter group:
 - catalytic rate constants,
 - Michaelis–Menten constants,
 - degradation rate constants.
- The minimum observation requirement can be identified based on a predefined parameter-error criterion.

- Determine which MAPK equations can be recovered most accurately using different machine learning approach.

References



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Thank You Questions?